

BULGAN, Mongolia

WMO: 442390

Lat: 48.8182N

Lon: 103.5190E

Elev: 3963

StdP: 12.71

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.5	(c) -24.4	(d) -34.3	(e) 0.9	(f) -27.4	(g) -29.8	(h) 1.2	(i) -23.6	(j) 19.5	(k) 10.4	(l) 16.5	(m) 5.8	(n) 3.6	(o) 290	(p) 0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.3	(c) 84.8	(d) 62.8	(e) 81.0	(f) 61.4	(g) 77.5	(h) 60.3	(i) 66.7	(j) 78.6	(k) 64.6	(l) 75.4	(m) 62.8	(n) 73.0	(o) 7.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	99.6	71.8	60.9	92.5	69.7	59.0	86.3	68.0	33.8	78.9	32.0	75.6	30.5	72.9	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	16.3	14.1	-32.2	90.8	5.5	4.5	-36.2	94.0	-39.4	96.6	-42.6	99.1	-46.6	102.4
			-32.2	70.4	5.4	4.6	-36.1	73.7	-39.3	76.4	-42.3	79.0	-46.3	82.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	-4.8	2.2	19.7	37.4	48.5	59.7	64.1	59.9	48.4	33.3	12.9	-0.4		
	DBStd	25.70	9.82	10.84	12.00	8.62	8.12	6.65	5.37	5.61	7.62	8.44	12.16	10.07		
	HDD50	7804	1697	1339	938	383	126	8	0	4	115	518	1114	1562		
	HDD65	12168	2162	1759	1403	827	511	181	80	174	497	983	1564	2027		
	CDD50	1201	0	0	0	6	81	298	437	311	68	0	0	0		
	CDD65	90	0	0	0	0	1	21	52	16	0	0	0	0		
	CDH74	1609	0	0	0	2	106	485	692	295	29	0	0	0		
	CDH80	422	0	0	0	0	18	122	209	70	3	0	0	0		
	WSAvg	5.5	4.1	4.6	5.8	6.9	7.2	6.4	5.5	5.3	5.8	5.3	4.5	4.4		
	PrecAvg	12.1	0.0	0.1	0.2	0.5	0.9	2.2	3.9	2.9	1.2	0.4	0.2	0.1		
	PrecMax	19.8	0.4	0.4	1.2	2.7	3.2	5.8	8.5	5.9	3.6	1.6	0.7	0.5		
(15) Precipitation	PrecMin	5.6	0.0	0.0	0.0	0.0	0.0	0.2	0.6	0.1	0.0	0.0	0.0	0.0		
	PrecStd	3.3	0.1	0.1	0.2	0.4	0.7	1.1	2.0	1.4	0.8	0.3	0.2	0.1		
	0.4%	DB	26.6	35.6	58.2	71.2	83.2	88.7	91.3	86.9	77.6	64.1	48.7	28.8		
	MCWB	22.4	29.0	41.9	49.6	56.7	61.7	67.2	67.2	63.5	57.3	47.9	38.4	24.2		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	19.3	30.0	49.8	64.7	77.3	83.5	85.7	80.9	72.7	58.3	42.2	22.7		
	MCWB	16.1	24.7	37.7	46.7	55.0	61.1	65.2	62.3	56.0	44.4	33.8	19.2	19.2		
	5%	DB	14.5	24.7	43.7	59.8	72.6	79.6	81.3	76.8	68.5	53.4	36.1	18.7		
	MCWB	12.2	20.8	34.1	44.2	52.8	59.7	63.5	61.0	53.6	41.3	30.0	16.1	16.1		
	10%	DB	10.4	19.5	38.6	55.3	68.1	75.8	77.3	72.9	63.9	48.5	30.6	14.4		
	MCWB	8.7	16.8	31.2	41.9	50.6	58.7	62.2	60.2	51.4	39.0	26.5	12.3	12.3		
	0.4%	WB	22.7	29.8	43.0	51.7	61.2	66.4	71.6	67.9	61.4	49.6	39.2	24.6		
	MCDB	26.3	34.9	56.4	66.6	77.6	82.1	86.4	79.0	71.4	61.8	47.9	28.6	28.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	16.4	24.9	38.2	47.9	57.3	63.9	68.3	65.3	57.9	45.2	33.9	19.7		
	MCDB	19.2	29.8	48.6	62.1	71.0	77.6	81.3	75.3	69.0	56.5	41.2	22.6	22.6		
	5%	WB	12.3	21.0	34.4	44.9	54.4	62.0	65.8	63.2	55.1	42.0	30.3	16.2		
	MCDB	14.3	24.7	43.1	58.4	69.0	75.1	76.7	72.8	66.2	52.2	35.4	18.8	18.8		
	10%	WB	8.6	16.8	31.2	42.3	51.6	60.2	63.8	61.2	52.6	39.2	26.5	12.4		
	MCDB	10.5	19.6	38.1	54.2	65.8	72.4	73.5	70.9	62.2	48.2	30.7	14.4	14.4		
	MDBR	20.4	25.6	27.1	25.9	27.4	24.8	21.3	22.3	26.1	25.3	21.8	19.0	19.0		
	5% DB	MCDBR	24.5	30.7	32.1	35.0	37.8	33.2	30.0	30.6	33.9	32.6	26.7	22.6		
(36) Mean Daily Temperature Range	MCWBR	21.7	26.0	21.4	19.9	19.4	15.3	13.3	14.5	18.4	20.0	19.4	19.9	19.9		
	5% WB	MCDBR	24.2	30.4	30.8	32.7	33.4	28.0	24.8	25.4	30.0	31.4	25.8	22.6		
	MCWBR	21.6	25.8	21.0	19.9	19.1	14.7	13.0	14.0	17.4	20.1	19.3	20.0	20.0		
	taub	0.241	0.264	0.310	0.351	0.381	0.386	0.374	0.349	0.307	0.286	0.262	0.243	0.243		
(40) Clear-Sky Solar Irradiance	taud	2.337	2.311	2.262	2.260	2.208	2.299	2.383	2.429	2.491	2.475	2.384	2.315	2.315		
	Ebn at Noon	268	285	286	284	278	277	279	282	286	275	256	249	249		
	Edh at Noon	21	28	35	39	43	40	36	33	28	24	20	19	19		
	RadAvg	562	918	1367	1735	2034	2062	1946	1702	1428	957	584	436	436		
	RadStd	27	47	55	82	106	136	101	91	61	39	29	26	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	-4.93	N/A	-1.93	-2.30	N/A	N/A	N/A	N/A
	N/A	N/A	-3.38	N/A	-1.21	-1.42	N/A	N/A	N/A	N/A

Nomenclature: See separate page